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FINAL TERM PREPARATION

CS401 – ASSEMBLY LANGUAGE PROGRAMMING

100% CORRECT SOLVED REPEATED QUESTIONS FROM PAST PAPERS (2022 – 2024) BY STUDY BUDDY HUB

How does processor handle the multiple pins using single INT pin?

A processor handles multiple interrupt sources using a single INT pin by employing an interrupt controller. The controller manages and prioritizes multiple interrupt requests, sending a signal to the processor INT pin only when an interrupt needs to be serviced. This allows the processor to handle various interrupt sources efficiently through a single interrupt line.

What is the purpose of BIOS parameter block?

The BIOS parameter block (BPB) is a structure that provides parameters about the storage medium. It is stored in the first sector or the boot sector of the device. It is crucial for defining how the disk is organized and ensuring that BIOS can correctly read from and write to the disk.

Write the purpose of multi-threading?

The purpose of multi-threading is to improve the efficiency and performance of a program by allowing it to perform multiple tasks concurrently. This can lead to better resource utilization, increased responsiveness, and faster execution of complex tasks, especially on multi-core processors where threads can run in parallel.

Write names of VESA services?

VESA stands for Video Electronics Standards Association. It includes some important services such as:

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INT 10 – VESA – Get SuperVGA Information

INT 10 – VESA – Get SuperVGA Mode Information

INT 10 – VESA – Set VESA Video Mode

Difference b/w multi-tasking and multi-threading?

Characteristics	**Multi-tasking**	**Multi-threading**
Definition	Running multiple processes simultaneously.	Running multiple threads within a single process.
Scope	Involves multiple processes.	Involves multiple threads within a single process.
Resource Sharing	Processes are isolated and do not share memory directly.	Threads share the same memory space and resources.
Performance	Performance can be affected by the need for context switching and managing multiple processes.	Typically, more efficient for tasks that can be parallelized within a single process.
Management	Managed by the operating system.	Managed within the application or by a thread library or runtime system.

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Which interrupt is used for extended read operation on disk? Also mention the corresponding service number?

Interrupt and Service Number for Extended Disk Read Operation is as follows:

Interrupt: INT 13h

Service Number: 42h

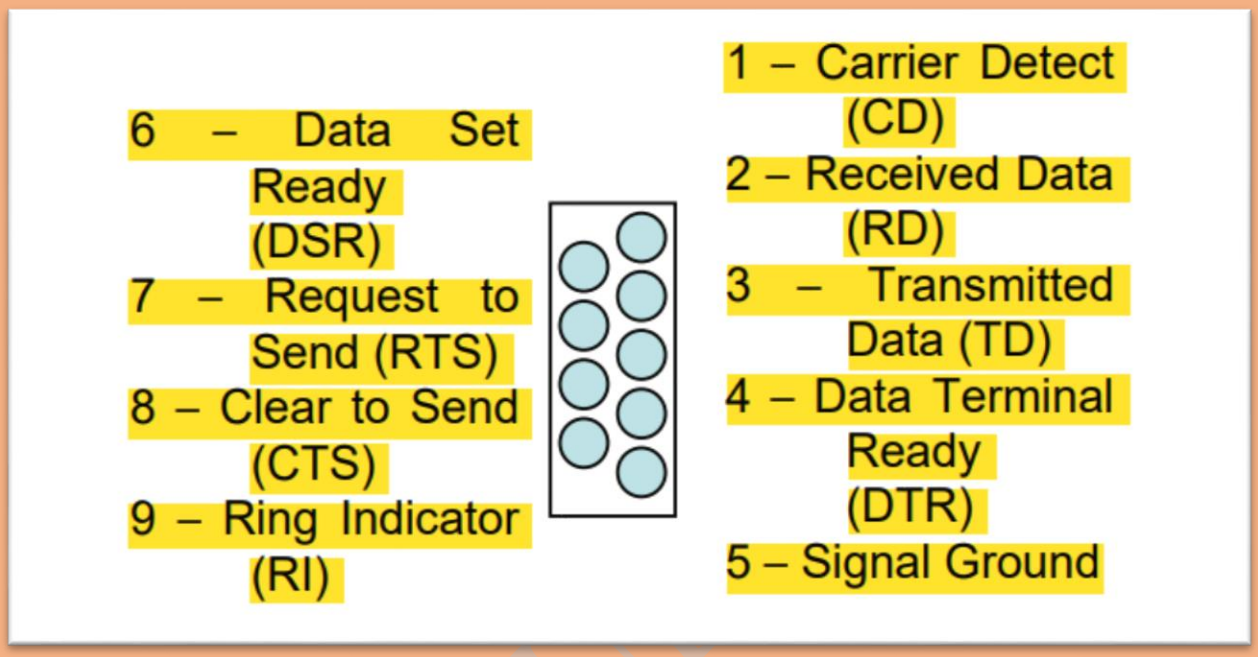
The interrupt INT 13h is used for disk operations, and service number 42h specifically handles extended read operations on the disk.

Difference b/w serial and parallel ports?

Characteristics	Serial Port	Parallel Port
Data Transmission	The bits are sent one by one on the serial port in a specially formatted fashion. It transmits data sequentially.	In case of parallel port, whole bytes are sent from one place to another. It transmits multiple bits simultaneously.
Number of wires	Less wires (typically 2 for data, plus control lines).	More wires (often 25 or more) for data and control.
Distance and interference	Better for long distances with less interference.	Better for short distances, more prone to signal degradation and crosstalk.

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Serial port is a way of communication among two devices just like the parallel port. The serial port connection is a 9pin DB-9 connector with pins assigned as shown below:



The data on the serial port is sent in which format? Also mention the baud rate of communication.

The data on the serial port is sent in standard format called RS232 communication. The duration of a bit must be precisely known at both ends called the baud rate of the communication.

What is the main reason of writing the TSR program in assembly language?

The main reason of writing the TSR (Terminate and Stay Resident) program is efficiency and Control. Assembly language provides low-level control over hardware and system resources, allowing TSR programs to run with minimal overhead and optimal performance. This is crucial for TSR programs, which need

to manage memory efficiently and execute quickly to remain resident in memory and respond to interrupts or events without significant delays.

Is it possible to print a string on the screen using BIOS interrupt 10?

Yes, it is possible to print a string on the screen using BIOS interrupt 10 (INT 10 service 13). BIOS will work even if the video memory is not at B8000 (a very old video card) since BIOS knows everything about the hardware and is hardware specific.

DOS uses a special service for free space allocation while loading a new program. Describe this special service and its procedure?

DOS uses the INT 21h interrupt with the function number 4Ah in the AH register for checking and allocating free space on a disk drive.

4Ah Parameters used are as follows:

AL: Drive number (0 for A, 1 for B)

BX: Number of sectors to allocate.

ES: DI: Address of a disk free structure (where space information is returned).

Assembly Code Example is as follows:

```
mov ah, 4Ah     ; Function: Check Disk Free Space
mov al, 0       ; Drive number (0 = A)
int 21h         ; Calling DOS interrupt
; After this call, AX will contain the number of free clusters
```

What B, D, and A represent in descriptor table.

The **Interrupt Descriptor table (IDT)** is not used for memory segmentation because it focuses on interrupt handling as compared to the GDT and LDT which

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are used for defining and managing memory segments. In the x86 architecture, the segment descriptor tables known as **Global Descriptor Table (GDT)** and **local descriptor table (LDT)**.

The **table index (TI)** is set to 0 to access the GDT (Global Descriptor Table). RPL is the requested privilege level that ranges from 0-3 and informs what privilege level the program wants when using this descriptor. The 13bit index is the actual index into the GDT to select the appropriate descriptor. 13 bits mean that a maximum of 8192 descriptors are possible in the GDT. The GDT itself is an array of descriptors where each descriptor is an 8-byte entry. The base and limit of GDT is stored in a 48bit register called the **GDTR**.

B (Busy Bit): Not typically used in GDT/LDT segment descriptors. It is relevant in Task State Segments (TSS) to indicate if a task is currently being used.

D (Default Operation Size): Indicates whether the segment uses 16-bit or 32-bit data. The 16-bit operations are represented by 0 and 32-bit operations are by 1.

A (Accessed Bit): Shows if the segment has been accessed (read or executed) since the last reset where 0 is for indicating the segment has not accessed and 1 is for indicating the segment has been accessed.

Write an assembly language program to display your name on the top left corner of screen using BIOS interrupt 10.

Assembly language program to display "Aroosa" at the top-left corner of the screen with light gray text on a black background is as follows:

```
BITS 16          ; Use 16-bit mode
[ORG 0x0100]          ; Origin for .COM files

; Set the video mode to 80x25 color text mode (mode 03h)
```

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```
MOV AH, 00h       ; BIOS function to set video mode
MOV AL, 03h       ; Mode 03h: 80x25 color text mode
INT 10h           ; Call BIOS interrupt

; Set the screen attributes: light gray text on black background
MOV AH, 0x06              ; BIOS function to set the text attribute for the whole screen
MOV AL, 07h       ; Text attribute (light gray on black)
MOV BH, 0         ; Page number (0 for default)
MOV CX, 0         ; Starting position (top-left corner)
MOV DX, 184Fh     ; Ending position (bottom-right corner)
INT 10h           ; Call BIOS interrupt

; Set the cursor position to (0,0)
MOV AH, 02h       ; BIOS function to set cursor position
MOV BH, 0         ; Page number (0 for default)
MOV DH, 0         ; Row (0 = top row)
MOV DL, 0         ; Column (0 = left column)
INT 10h           ; Call BIOS interrupt

; Set up for displaying characters
MOV AH, 0Eh               ; BIOS teletype function (TTY) to display characters
MOV BH, 0         ; Page number (0 for default)
MOV BL, 7         ; Text attribute (light gray on black)

; Define the string to be printed
```

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```
MOV SI, string     ; Load address of the string into SI
print_char:
LODSB              ; Load byte at DS:SI into AL and increment SI
CMP AL, '$'        ; Check for string terminator '$'
JE done            ; If terminator, jump to done
MOV DL, AL         ; Move character from AL to DL for printing
INT 10h            ; Call BIOS interrupt
JMP print_char     ; Repeat for next character
done:
MOV AH, 4Ch        ; Terminate program function
INT 21h            ; Call DOS interrupt to exit
; Define the string to print
string DB 'A', 'r', 'o', 'o', 's', 'a', '$'
END                ; End of the program
```

What is PCB? Also write the instruction used for restoring the registers?

The space where all registers of a task are stored is called the **process control block or PCB**. The instruction INT 08 that is saving and restoring the registers is called the scheduler and the whole event is called a **context switch**.

Explain Dos input buffer?

The DOS input buffer is a temporary storage area in memory used to hold user input characters before they are processed by a program. It is used by DOS interrupt functions to capture and store input data from the keyboard. The buffer is managed by DOS and is typically accessed via interrupts, such as INT 21h, which provide various services for reading input.

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The function INT 21h, AH = 0Ah reads a string of characters from the keyboard into a predefined buffer. The buffer format includes:

Maximum Length: Specifies the maximum number of characters the buffer can hold.

Actual Length: Indicates the number of characters read into the buffer.

String Data: The actual user input string, including a carriage return (CR) if entered.

The function INT 21h, AH = 01h reads a single character from the keyboard.

The buffer allows programs to handle user input efficiently and can be manipulated directly for tasks such as text processing or command handling.

How do you recognize the zero in string value.

To recognize a zero (null terminator) in a string in assembly language programming, we need to:

Check Each Byte: Use CMP to compare each byte in the string with 0x00.

Example: CMP AL, 0x00 where AL holds the current byte.

Terminate on Zero: Use JE (Jump if Equal) to branch to a section of code when the null terminator is found.

Example: JE end_of_string tells the processor to jump to the end_of_string label when it finds the null terminator in the string. This allows you to handle the end of the string appropriately, such as processing the string or terminating the program.

Explain the purpose of "JMP $" instruction?

JMP $ Instruction: The JMP $ instruction causes the program to jump to its own address. The $ symbol represents the current address of the instruction.

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Purpose: It creates an infinite loop. When the CPU executes JMP $, it continuously jumps to the same instruction, effectively halting further progress in the program and looping indefinitely.

Which registers must be preserved?

Both standards (C and Pascal) preserve EBX, ESI, EDI, EBP, ESP, DS, ES, and SS registers.

Write the instruction for returning EOI Signal?

The End of Interrupt (EOI) signal is used in the context of interrupt handling, particularly with the Programmable Interrupt Controller (PIC). To signal the PIC that the interrupt has been handled and to allow the PIC to send new interrupts, the EOI signal must be sent.

Which architecture is also called load-store architecture?

SPARC stands for Scalable Processor architecture, a 64bit processor which is a totally register based architecture, or it is also called a load-store architecture where memory access is only allowed in data movement instruction.

What is the naming convention?

C prepends an underscore to every function or variable name while Pascal translates the name to all uppercase. C++ has a weird name mangling scheme that is compiler dependent. To avoid it C++ can be forced to use C style naming with extern “C” directive.

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Write the instruction for taking input from the keyboard port?

To read a character from the keyboard port, we use the I/O port 0x60. This port is used for keyboard input in x86 architecture.

```
MOV AL, 60h          ; Load the port number (0x60) into AL

IN AL, 60h           ; Read a byte from the keyboard data port (0x60) into AL
```

SPARC instructions have two sources and a distinct destination. This allows more flexibility in writing programs. Some examples of instructions of this processor follow.

```
Data Movement
LDSB [rn], rn                (load signed byte)
LDUW [rn], rn                (load unsigned word)
STH [rn], rn                 (store half word)
Arithmetic
source1 = rn
source2 = rn or simm13
dest = rn
ADD r2, r3, r4
SUB r2, 4000, r5
SLL, SRA, SRL                (shifting)
AND, OR, XOR                 (logical)
Program Control
CALL                         (direct call)
JMPL                         (register indirect)
RET
SAVE
RESTORE
BA label                     (Branch Always)
BE label                     (branch if equal)
BCC label                    (branch if carry clear)
BLE label                    (branch if less or equal)
```

Which registers are used as scratch?

Both standards (C and Pascal) do not preserve or guarantee the value of EAX, ECX, EDX, FS, GS, EFLAGS, and any other registers.

Which register holds the return value?

Both C and Pascal return up to 32bit large values in EAX and up to 64bit large values in EDX: EAX.

Who is responsible for removing the parameters?

In C the caller removes the parameter while in Pascal the callee removes them. The C scheme has reasons pertaining to its provision for variable number of arguments.

What the given instructions are showing?

XOR AX, F0

OR BX, 0F

AND BX, F0

AND DX, 0

These instructions perform bitwise operations to modify specific bits or clear registers as needed.

- **XOR AX, F0**:

Operation: Performs a bitwise XOR between the AX register and the hexadecimal value 0xF0.

Effect: Toggles (flips) the upper 4 bits (4-7) of AX. The lower 4 bits (0-3) remain unchanged.

Example:

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If AX is 0x3F (0011 1111 in binary) and 0xF0 is (1111 0000 in binary), then AX XOR 0xF0 will result in 0xCF (1100 1111 in binary).

- **OR BX, 0F**:

Operation: Performs a bitwise OR between the BX register and the hexadecimal value 0x0F.

Effect: Sets the lower 4 bits (0-3) of BX to 1. The upper bits (4-15) remain unchanged.

Example:

If BX is 0xA5 (1010 0101 in binary) and 0x0F is 0000 1111 in binary, then BX OR 0x0F will result in 0xAF (1010 1111 in binary).

- **AND BX, F0**:

Operation: Performs a bitwise AND between the BX register and the hexadecimal value 0xF0.

Effect: Clears the lower 4 bits (bits 0-3) of BX and retains the upper 4 bits (bits 4-15) unchanged.

Example:

If BX is 0xAB (1010 1011 in binary) and 0xF0 is 1111 0000 in binary, then BX AND 0xF0 will result in 0xAA (1010 0000 in binary).

- **AND DX, 0**:

Operation: Performs a bitwise AND between the DX register and 0x00.

Effect: Clears all bits in DX, effectively setting DX to 0. This is commonly used to reset or clear a register.

Example:

Any value in DX AND with 0x00 will result in 0x00.

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What is REP instruction in assembly language?

REP stands for Repeat Prefix. The purpose of REP is that it repeats a string instruction (like MOVSB, LODSB, STOSB) until the CX register reaches zero. The instruction REP is executed multiple times based on the CX register value.

How are parameters passed to the routine?

In C, the parameters are pushed in reverse order with the rightmost being pushed first. In Pascal, the parameters are pushed in proper order with the leftmost being pushed first.

What is RET instruction in assembly language?

RET means return. This instruction is used to return from a procedure or function. It pops the return address from the stack and transfers control to that address.

Mention the values of AX, AH, AL, BX, CX, DX registers CF when we "SET CURRENT FILE POSITION" using INT 21?

The values of registers when using INT 21h to set the file position with function code 0x42 are as follows:

AX: Generally, not directly affected, but AX may contain status or result based on the function.

AL: Typically, not used for this function.

AH: Contains the function code (for setting file position, AH = 42h).

BX: Contains the file handle.

CX: Contains the high word of the file position to set.

DX: Contains the low word of the file position to set.

CF: Carry Flag is set if there is an error; cleared if the operation is successful.

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Difference b/w parameter passing in Pascal and C?

Characteristics	Parameter passing in Pascal	Parameter passing in C
Default parameter passing	By default, Pascal uses "pass-by-value" for basic data types and "pass-by-reference" for arrays and records. In Pascal, "pass-by-value" is used for simple types and "pass-by-reference" is for complex types.	By default, C uses "pass-by-value". In C, pass-by-value is used for all types, with pointers used to simulate pass-by-reference.
Pass-by-Value	Copies the actual value of the parameter into the function or procedure.	Copies the value of the parameter to the function. Changes to the parameter inside the function do not affect the original variable.
Pass-by-Reference	Passes a reference (address) to the actual parameter, allowing the function to modify the original variable.	Simulated using pointers. We pass the address of the variable, allowing the function to modify the original variable.

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Which has the same meaning as in data and code descriptors?

The P and DPL have the same meaning as in data and code descriptors. The S bit tells that this is a system descriptor while the 1110 following it tells that it is a 386-interrupt gate.

Explain VESA?

VESA is the Video Electronics Standards Association and VBE is the set of Video BIOS Extensions proposed by them. The VESA VBE 2.0 standard includes a linear frame buffer mode. This mode allows direct access to the whole video memory.

Which system can define the possible descriptors for a program and the program is bound to select one of them and nothing else. Explain this sentence?

Operating system can define the possible descriptors for a program and the program is bound to select one of them and nothing else. This sentence also hints that the processor has some sense of programs that can or cannot do certain things like change this table of descriptors. This is called the privilege level of the program and varies for 0 (highest privilege) to 3 (lowest privilege).

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